

Random Variable & Mathematical Expectation

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Random Variable

A random variable is a variable that takes on **numerical values** as a result of a random experiment or measurement; associates a numerical value with each possible outcome.

The differences between variable and random variable are-

- Random variable always takes **numerical values**
- There is a **probability associated** with each possible values

Random variable is denoted by capital letters such as X, Y, Z etc.

And the possible outcomes are denoted by small letters such as x, y, z etc.

Random Variable

Example 1:

A coin is tossed. It has two possible outcomes- Head and Tail.

Consider a variable, $X = \text{outcome of a coin toss} = \begin{cases} H, & \text{if Head appears} \\ T, & \text{if Tail appears} \end{cases}$

Here, $S = \{H, T\}$.

But, these are not numerical values.

Random Variable

Example 1(contd.):

Consider a variable, X = Number of heads obtained in a trial

$$\text{Then, } X = \begin{cases} 1, & \text{if Head appears} \\ 0, & \text{if Tail appears} \end{cases}$$

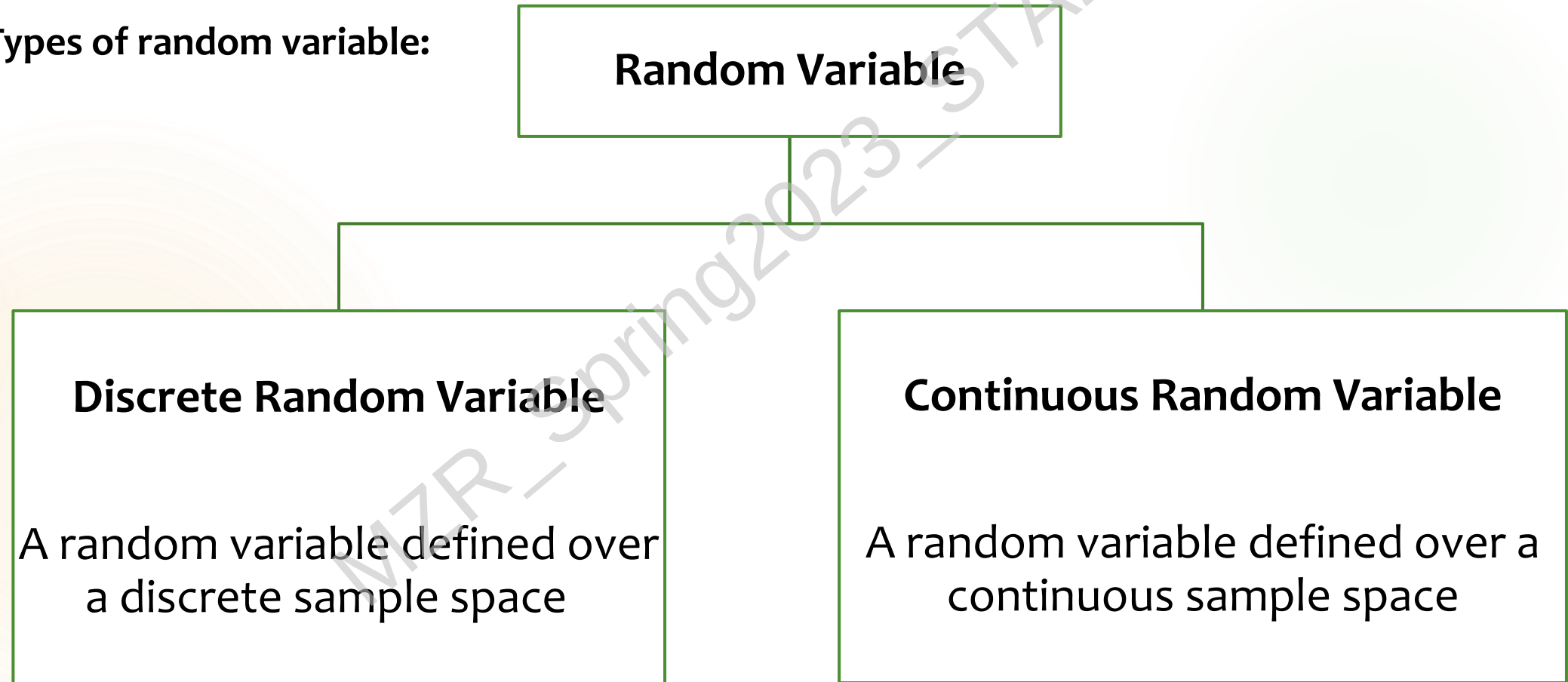
For a fair coin, we can write, $P(X=1) = \frac{1}{2}$ and $P(X=0) = \frac{1}{2}$

So, X is a random variable.

Random Variable

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Types of random variable:



Random Variable

Examples:

Discrete Random Variable:

1. X = Number of correct answers in a 100-MCQ test = 0, 1, 2, ..., 100
2. X = Number of cars passing a toll both in a day = 0, 1, 2, ..., ∞
3. X = Number of balls required to take the first wicket = 1, 2, 3, ..., ∞
4. X = The number of telephone calls received in a telephone booth during one day = 1, 2, ...

Continuous Random Variable:

1. X = Weight of a person. $0 < X < \infty$
2. X = Monthly Profit. $-\infty < X < \infty$
3. X = Temperature recorded by the meteorological office. $0 < X < \infty$

Probability Distributions

Distribution of the probabilities among the different values of a random variable.

Discrete probability distribution- probability distribution of a discrete random variable

Continuous probability distribution- probability distribution of a continuous random variable

Probability Distributions

Examples:

Discrete probability distribution-

- Tossing a coin 2 times.

X = Number of Heads appeared

$S = \{HH, HT, TH, TT\}$

x	0	1	2
$P(x)$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{1}{4}$

Probability Distributions

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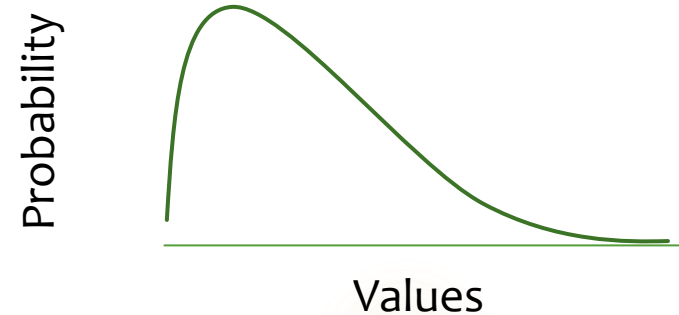
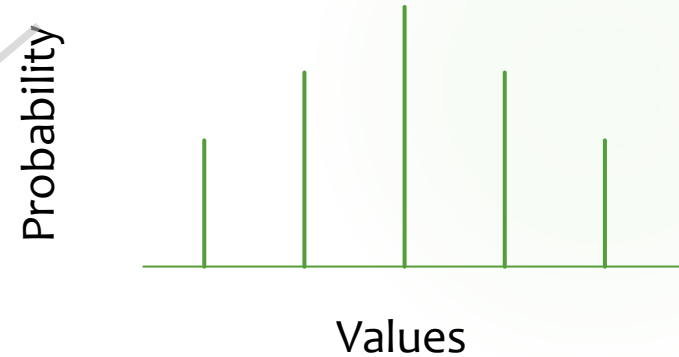
Different types of probability distributions:

Discrete probability distribution-

1. Bernoulli Distribution
2. *Binomial Distribution*
3. Poisson Distribution etc.

Continuous probability distribution-

1. Uniform Distribution
2. *Normal Distribution*
3. Exponential Distribution
4. t-distribution etc.



PMF and PDF

Probability Mass Function (pmf)- the probability distribution function of a discrete random variable X is called a pmf and is denoted by $p(x)$

Properties of probability function:

If $p(x)$ is probability function of a discrete random variable X , then $p(x)$ satisfies the following two properties:

- ▶ $0 \leq p(x) \leq 1$, For each possible value of X ,
- ▶ $\sum p(x_i) = 1$

Probability Density Function (pdf)- the probability distribution function of a continuous random variable X is called a pdf and is denoted by $f(x)$

If $f(x)$ is probability function of a discrete random variable X , then $f(x)$ satisfies the following two properties:

1. $f(x) = 0$
2. $\int_{-\infty}^{\infty} f(x) dx = 1$
3. $P[a \leq x \leq b] = \int_a^b f(x) dx$

PMF

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Example:

Let X be a random variable with probability function defined as follows

Values of $X : x$	- 2	0	4	11
$f(x)$	1/10	2/10	4/10	3/10

Find:

- i. $P[-2 \leq x < 4]$ ii. $P[x > 0]$ iii. $P[x \leq 4]$

Answer:

- i. $P[-2 \leq x < 4] = P[X = -2] + P[X = 0] = \dots \dots \dots =$
- ii. $P[x > 0] = \dots \dots \dots = \dots \dots \dots =$
- iii. $P[x \leq 4] = \dots \dots \dots = \dots \dots \dots =$

PMF

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Problem:

A random variable X has the following probability function:

X Values of x	0	1	2	3	4	5	6	7	8
$f(x)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

- Determine the value of a .
- Find $P[x < 3]$, $P[x \geq 3]$ and $P[0 < x < 5]$

Problem:

A coin is tossed three times in which the probability of head is twice as the probability of tail. If the number of heads is a random variable, find the probability function of the random variable. Also find

a. $P[x \geq 1]$

b. $P[x = 2]$

c. $P[x \leq 1]$

Mathematical Expectations

- For a discrete random variable X with pmf $p(x)$, the mathematical expectation of X is-

$$\mu = E(X) = \sum_x x p(x)$$

- For a continuous random variable X with pdf $f(x)$, the mathematical expectation of X is-

$$\mu = E(X) = \int_x x f(x)$$

Mathematical expectation is also known as population mean or expected value.

Mathematical Expectations

$$E(X^2) = \begin{cases} \sum_x x^2 p(x) & , \text{if } x \text{ is a discrete r.v.} \\ \int_x x^2 f(x) & , \text{if } x \text{ is a continuous r.v.} \end{cases}$$

Variance:

$$\sigma^2 = \text{Var}(X) = E[X - E(X)]^2 = E(X^2) - [E(X)]^2 = E(X^2) - \mu^2$$

Standard deviation: $\sigma = \sqrt{\text{Var}(X)}$

Properties of Mathematical Expectations

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Let, c is a constant number

X and Y are two independent random variables

1. $E(c) = c$

2. $E(cX) = c E(X)$

3. $E(X + c) = E(X) + c$

4. $E(X+Y) = E(X) + E(Y)$

5. $E(X-Y) = E(X) - E(Y)$

6. $E(XY) = E(X) \cdot E(Y)$

1. $\text{Var}(c) = 0$

2. $\text{Var}(cX) = c^2 \text{Var}(X)$

3. $\text{Var}(X + c) = \text{Var}(X)$

4. $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$

5. $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$

Mathematical Expectation

Example 2-

A company estimates the net profit on a new product, it is launching, to be Rs. 3 million during first year, if it is 'successful', Rs. 1 million if it is 'moderately successful', and a loss of Rs. 1 million if it is 'unsuccessful'.

The company assigns the following probabilities to first year prospects for the product-

Successful: 0.25, Moderately successful: 0.40, and Unsuccessful: 0.35

What are the **expected value** and **standard deviation** of the first year net profit for the product? Also, find the expected value of net profit if there is a fixed cost of Rs. 0.2 million, whatever the success status is.

Mathematical Expectation

Solution-

Let,

X = Net profit on the new product in the 1st year (Rs. Million)

Given that,

x	3	1	-1
$P(x)$	0.25	0.4	0.35

$$\text{Expected net profit, } E(X) = \sum x p(x) = (3 * 0.25) + (1 * 0.4) + (-1 * 0.35) \\ = 0.8 \text{ million}$$

Mathematical Expectation

Solution (contd.)-

$$\begin{aligned} E(X^2) &= \sum x^2 p(x) = (3^2 * 0.25) + (1^2 * 0.4) + ((-1)^2 * 0.35) \\ &= (9 * 0.25) + (1 * 0.4) + (1 * 0.35) = 3 \end{aligned}$$

$$\begin{aligned} Var(X) &= E(X^2) - [E(X)]^2 = 3 - 0.8^2 = 2.36 \\ \therefore SD(X) &= \sqrt{Var(X)} = \sqrt{2.36} = 1.54 \text{ million} \end{aligned}$$

If there is a fixed cost of Rs. 0.2 million, then expected net profit-

$$E(X - 0.2) = E(X) - 0.2 = 0.8 - 0.2 = 0.6 \text{ million}$$

Mathematical Expectation

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Sometimes $E(X)$ is called as mathematical expectation of X or expected value of X or mean of the distribution.

Problem:

Find the mean of a random variable having probability function defined as follows:

Values of $X : x$	- 2	0	4	11
$f(x)$	1/10	2/10	4/10	3/10